

Virtualization-based Security (VBS)

02/27/2025 Applies to:  [Windows 11](#),  [Windows 10](#)

Virtualization-based security, or VBS, uses hardware virtualization and the Windows hypervisor to create an isolated virtual environment that becomes the root of trust of the OS that assumes the kernel can be compromised. Windows uses this isolated environment to host a number of security solutions, providing them with greatly increased protection from vulnerabilities in the operating system, and preventing the use of malicious exploits which attempt to defeat protections. VBS enforces restrictions to protect vital system and operating system resources, or to protect security assets such as authenticated user credentials.

One such example security solution is [memory integrity](#), which protects and hardens Windows by running kernel mode code integrity within the isolated virtual environment of VBS. Kernel mode code integrity is the Windows process that checks all kernel mode drivers and binaries before they're started, and prevents unsigned or untrusted drivers or system files from being loaded into system memory. Memory integrity also restricts kernel memory allocations that could be used to compromise the system, ensuring that kernel memory pages are only made executable after passing code integrity checks inside the secure runtime environment, and executable pages themselves are never writable. That way, even if there are vulnerabilities like a buffer overflow that allow malware to attempt to modify memory, executable code pages cannot be modified, and modified memory cannot be made executable.

Note

Memory integrity is sometimes referred to as *hypervisor-protected code integrity (HVCI)* or *hypervisor enforced code integrity*, and was originally released as part of *Device Guard*. Device Guard is no longer used except to locate memory integrity and VBS settings in Group Policy or the Windows registry.

VBS requires the following components be present and properly configured.

 [Expand table](#)

Hardware requirement	Details
64-bit CPU	Virtualization-based security (VBS) requires the Windows hypervisor, which is only supported on 64-bit IA processors with virtualization extensions, including Intel VT-X and AMD-v.
Second Level Address Translation (SLAT)	VBS also requires that the processor's virtualization support includes Second Level Address Translation (SLAT), either Intel VT-X2 with Extended Page Tables (EPT), or AMD-v with Rapid Virtualization Indexing (RVI).
IOMMUs or SMMUs (Intel VT-D, AMD-Vi, Arm64 SMMUs)	All I/O devices capable of DMA must be behind an IOMMU or SMMU. An IOMMU can be used to enhance system resiliency against memory attacks.
Trusted Platform Module (TPM) 2.0	For more information, see Trusted Platform Module (TPM) 2.0 .
Firmware support for SMM protection	System firmware must adhere to the recommendations for hardening SMM code described in the Windows SMM Security Mitigations Table (WSMT) specification . The WSMT specification contains details of an ACPI table that was created for use with Windows operating systems that support VBS features. Firmware must implement the protections described in the WSMT specification, and set the corresponding protection flags as described in the specification to report compliance with these requirements to the operating system.
Unified Extensible Firmware Interface (UEFI) Memory Reporting	UEFI firmware must adhere to the following memory map reporting format and memory allocation guidelines in order for firmware to ensure compatibility with VBS. UEFI v2.6 Memory Attributes Table (MAT) - To ensure compatibility with VBS, firmware must cleanly separate EFI runtime memory ranges for code and data, and report this to the operating system. Proper segregation and reporting of EFI runtime memory ranges allows VBS to apply the necessary page protections to EFI runtime services code pages within the VBS secure region. Conveying this information to the OS is accomplished using the EFI_MEMORY_ATTRIBUTES_TABLE. To implement the UEFI MAT, follow these guidelines: - The entire EFI runtime must be described by this table. - All appropriate attributes for EfiRuntimeServicesData and EfiRuntimeServicesCode pages must be marked. - These ranges must be aligned on page boundaries (4KB), and can not overlap. EFI Page Protections -All entries must include attributes EFI_MEMORY_RO, EFI_MEMORY_XP, or both. All UEFI memory that is marked executable must be read only. Memory marked writable must not be executable. Entries may not be left with neither of the attributes set, indicating memory that is both executable and writable.

Hardware requirement	Details
Secure Memory Overwrite Request (MOR) revision 2	Secure MOR v2 is enhanced to protect the MOR lock setting using a UEFI secure variable. This helps guard against advanced memory attacks. For details, see Secure MOR implementation .
Memory integrity-compatible drivers	Ensure all system drivers have been tested and verified to be compatible with memory integrity. The Windows Driver Kit and Driver Verifier contain tests for driver compatibility with memory integrity. There are three steps to verify driver compatibility: 1. Use Driver Verifier with the Code Integrity compatibility checks enabled. 2. Run the Hypervisor Code Integrity Readiness Test in the Windows HLK. 3. Test the driver on a system with VBS and memory integrity enabled. This step is imperative to validate the driver's behavior with memory integrity, as static code analysis tools simply aren't capable of detecting all memory integrity violations possible at runtime.
Secure Boot	Secure Boot must be enabled on devices leveraging VBS. For more information, see Secure Boot

VBS works on VMs that have either nested virtualization support or **Guest VSM** enabled. The latter is enabled by default for Generation 2 VMs on Hyper-V. This also includes all Generation 2 VMs on Microsoft Azure, and Generation 1 VMs that have nested virtualization enabled. A list of supported Azure VM series is detailed in the table below.

 Expand table

VM Series Name	Nested Virtualization	VM Generation
Av2	Yes	1 (certain internal sizes support generation 2)
B	No	1 and 2
Dsv2/Dv2/Dv3/Ev3	Yes	1
Dsv3/Ddsv3	Yes	1 and 2
Dsv4/Ddsv4	Yes	1 and 2
Esv3/Edsv3	Yes	1 and 2
Esv4/Edsv4	Yes	1 and 2

VM Series Name	Nested Virtualization	VM Generation
Ev4/Edv4	Yes	Ev4 - 1 only Edv4 -1&2
Dv4/Ddv4	Yes	1 and 2
Dv5/Ddv5/Dsv5/Ddsv5	Yes	1 and 2
Ev5/Edv5/Esv5/Edsv5	Yes	1 and 2
Dasv5/Dadsv5/Easv5/ Eadsv5	Yes	1 and 2
Ebsv5/Edbsv5	Yes	1 and 2
Fsv2	Yes	1 and 2
Fx	Yes	2
Lsv2	Yes	1 and 2

Related topics

For more info about Hyper-V, see [Hyper-V on Windows Server 2016](#) or [Introduction to Hyper-V on Windows 10](#). For more info about hypervisor, see [Hypervisor Specifications](#).